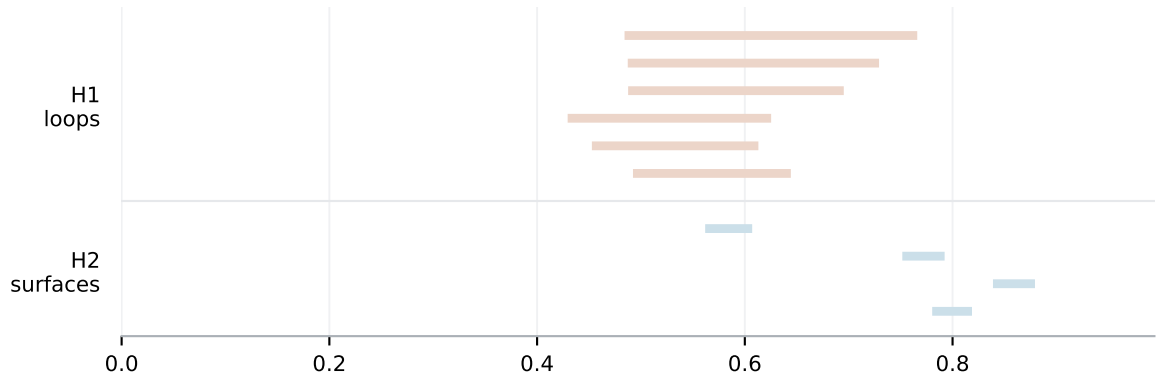
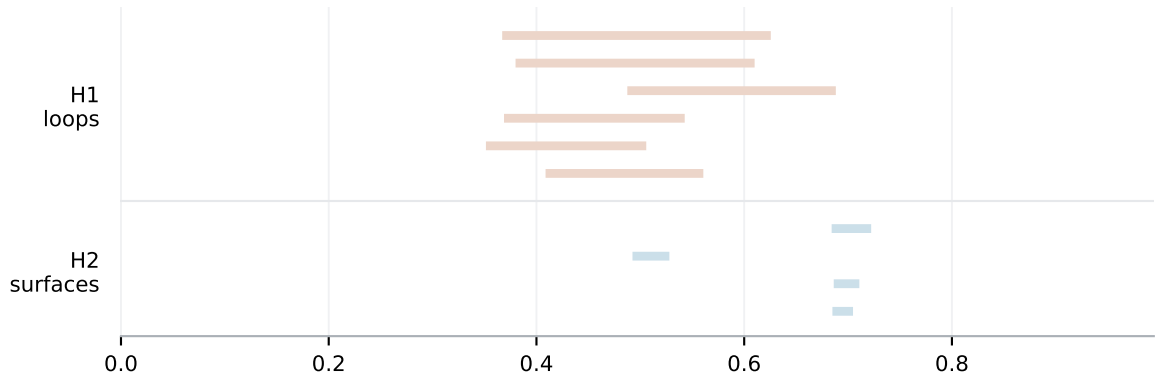
Persistence barcodes · all eight seed pairs
256 landmarks · original activity distances · no UMAP

Seed 1 · Grid only
Above null: 0 H1, 0 H2 · core: no; strict: no



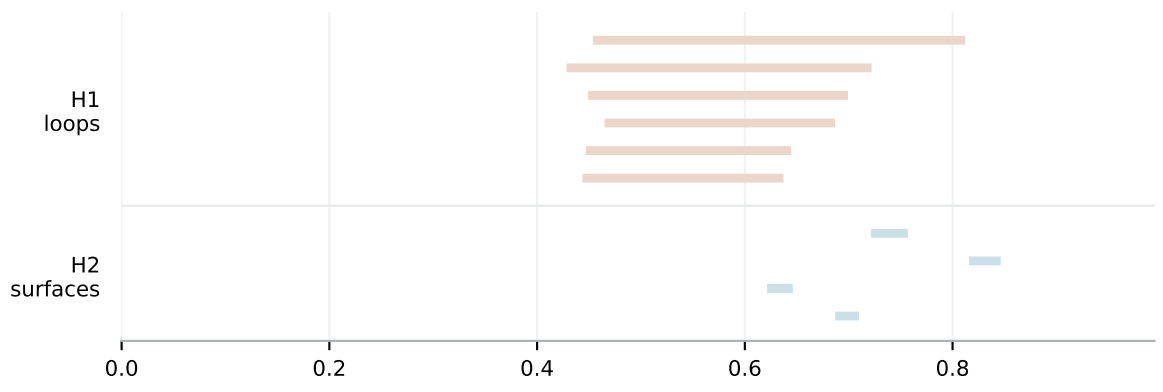
Seed 1 · Grid + 33% PGC
Above null: 0 H1, 0 H2 · core: no; strict: no



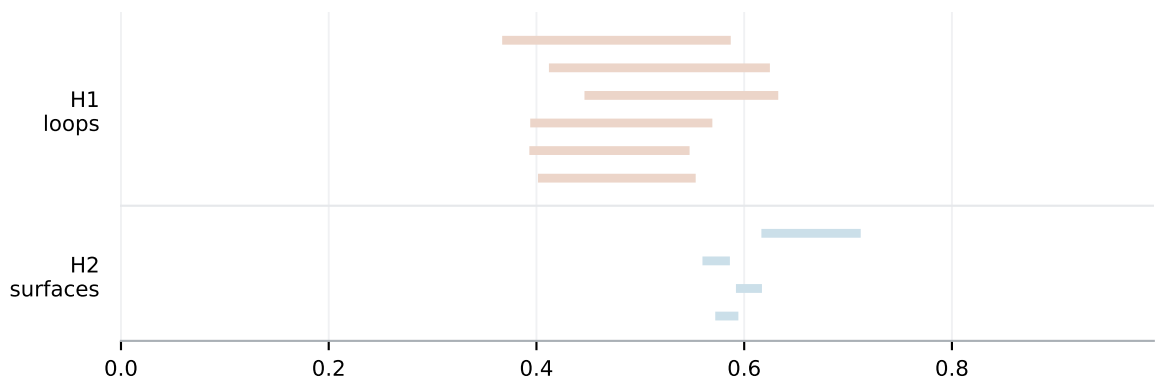
Seed 2 · Grid only
Above null: 0 H1, 0 H2 · core: no; strict: no

Seed 2 · Grid + 33% PGC
Above null: 0 H1, 0 H2 · core: no; strict: no

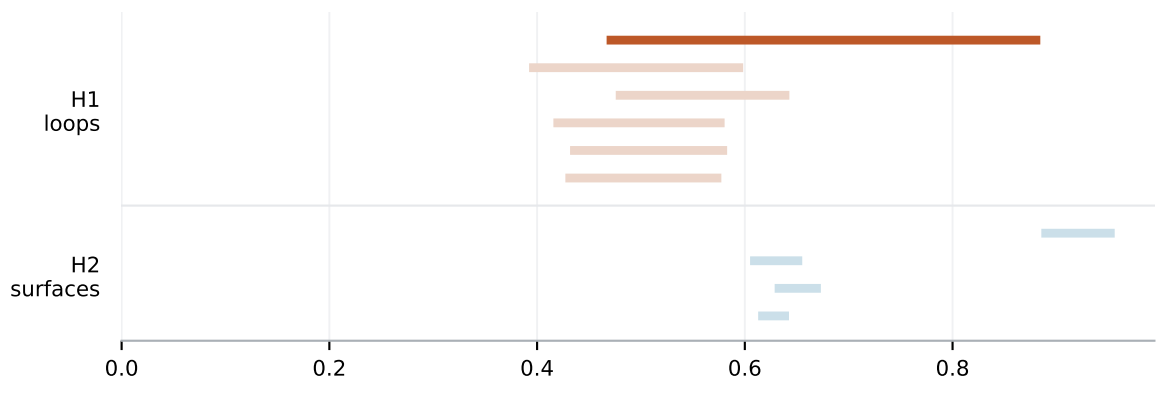
Seed 3 · Grid only
Above null: 0 H1, 0 H2 · core: no; strict: no



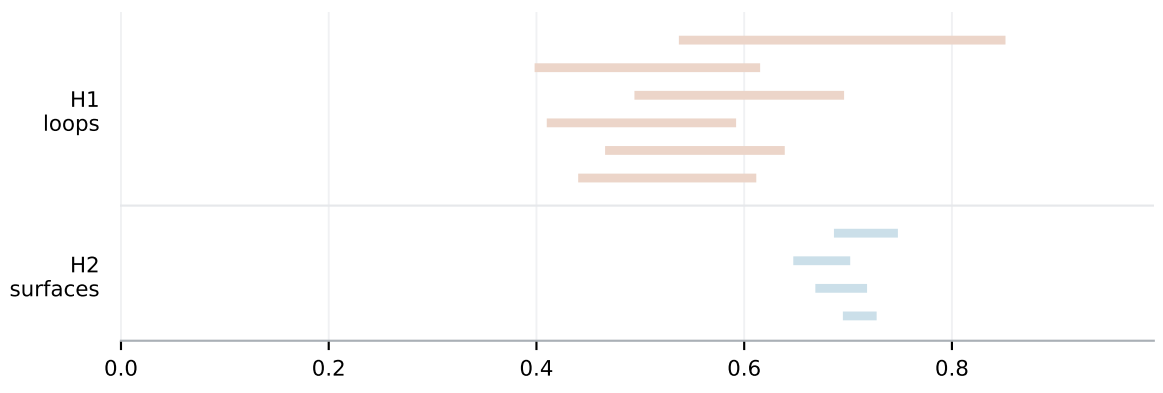
Seed 3 · Grid + 33% PGC
Above null: 0 H1, 0 H2 · core: no; strict: no



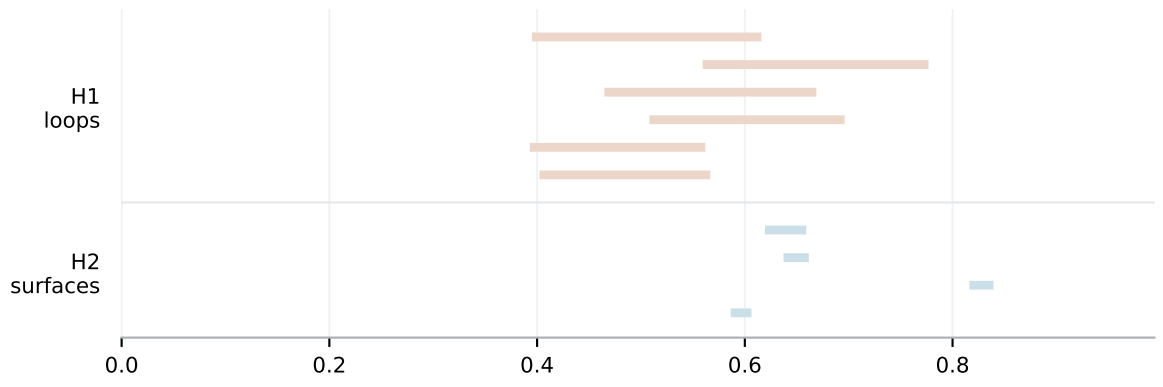
Seed 4 · Grid only
Above null: 1 H1, 0 H2 · core: no; strict: no



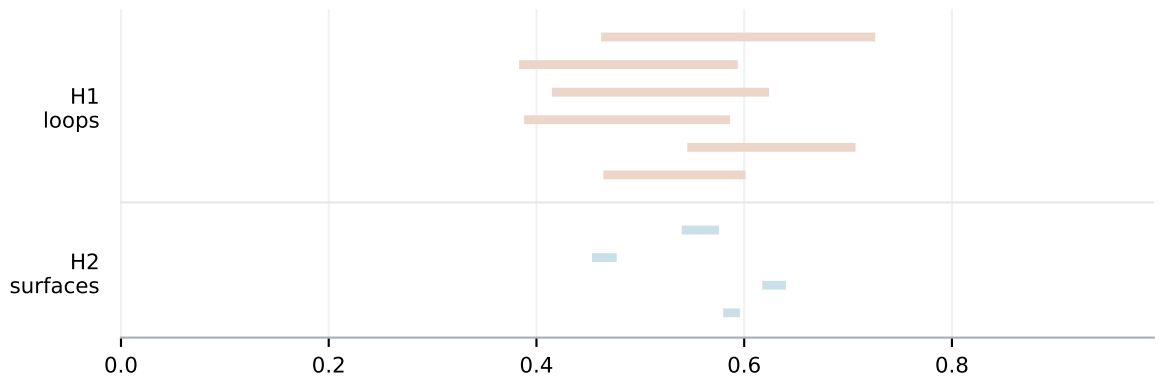
Seed 4 · Grid + 33% PGC
Above null: 0 H1, 0 H2 · core: no; strict: no



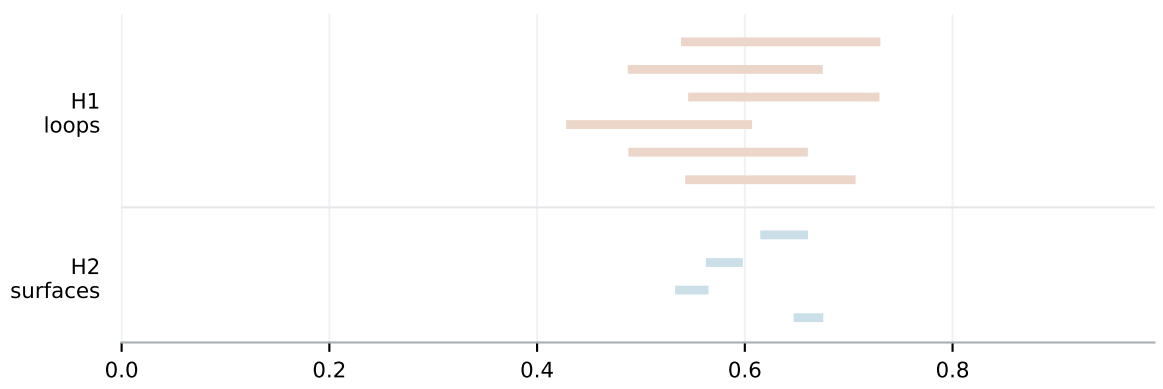
Seed 5 · Grid only
Above null: 0 H1, 0 H2 · core: no; strict: no



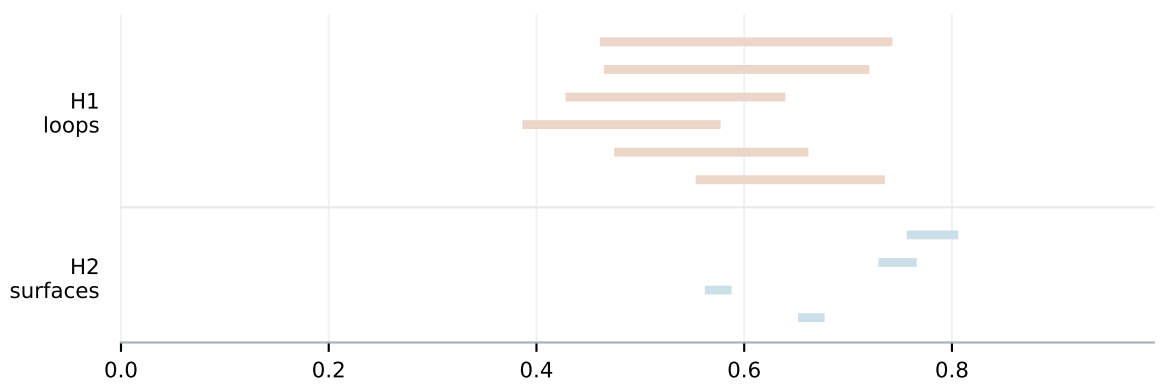
Seed 5 · Grid + 33% PGC
Above null: 0 H1, 0 H2 · core: no; strict: no



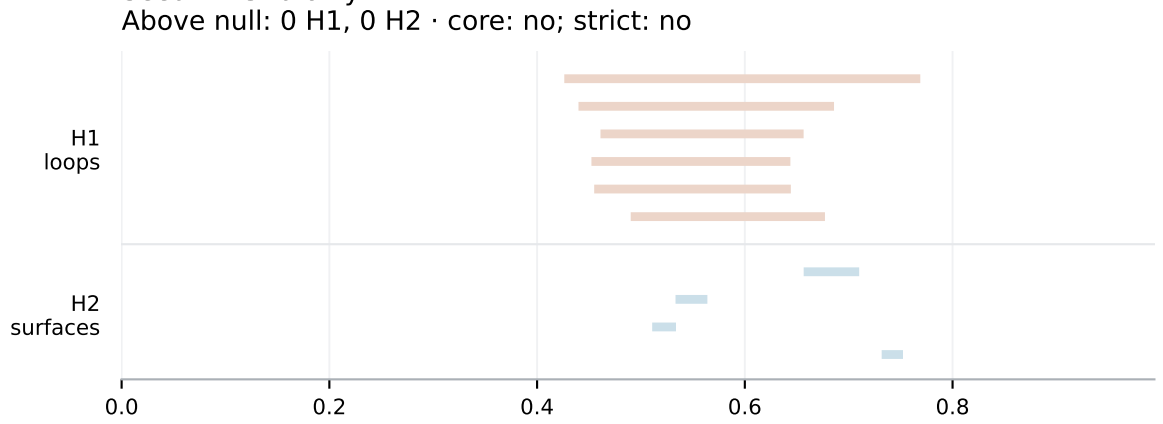
Seed 6 · Grid only
Above null: 0 H1, 0 H2 · core: no; strict: no



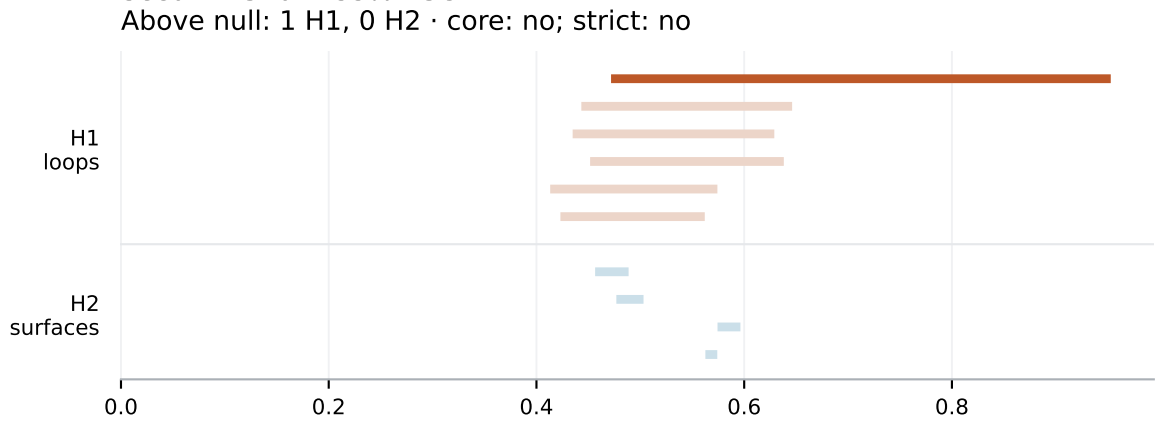
Seed 6 · Grid + 33% PGC
Above null: 0 H1, 0 H2 · core: no; strict: no



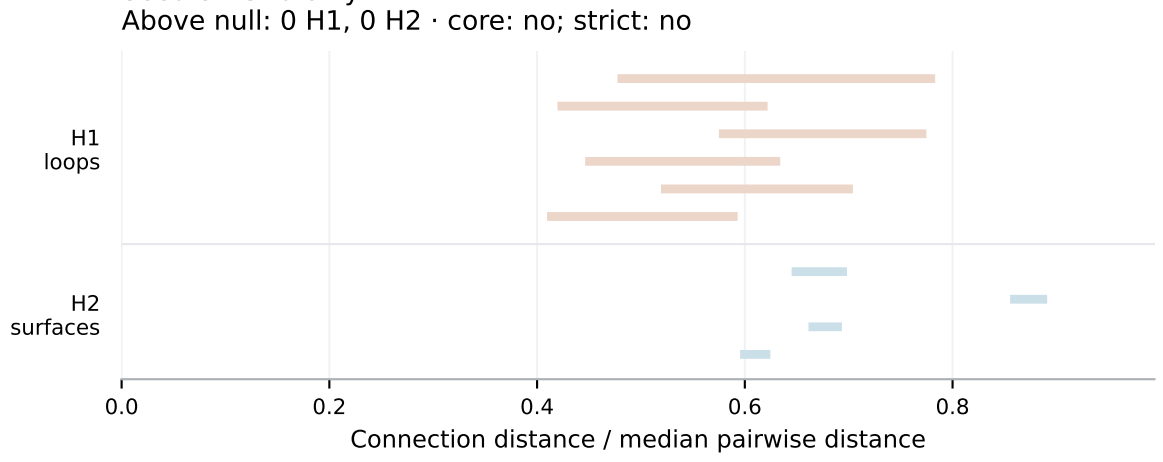
Seed 7 · Grid only
Above null: 0 H1, 0 H2 · core: no; strict: no



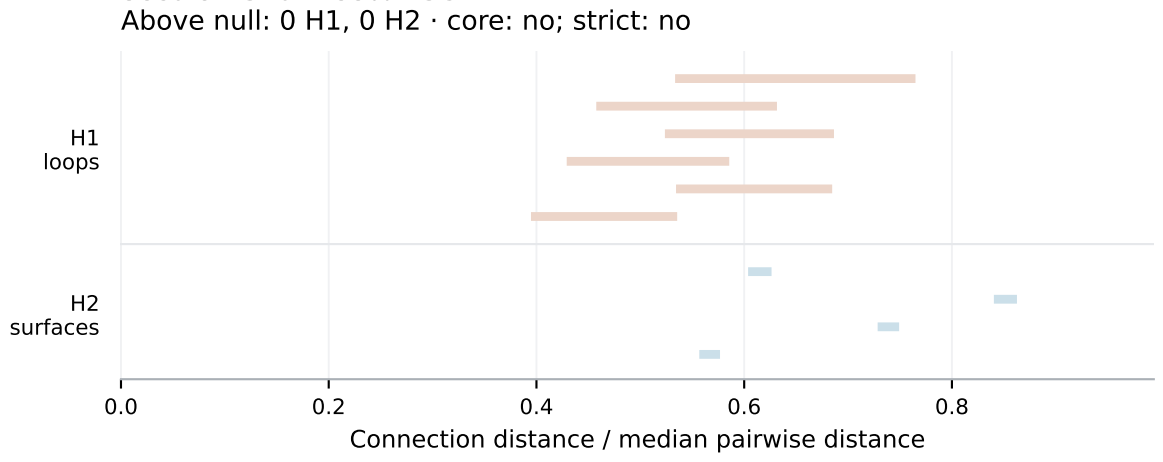
Seed 7 · Grid + 33% PGC
Above null: 1 H1, 0 H2 · core: no; strict: no



Seed 8 · Grid only
Above null: 0 H1, 0 H2 · core: no; strict: no



Seed 8 · Grid + 33% PGC
Above null: 0 H1, 0 H2 · core: no; strict: no



Each bar spans its birth and death distances; longer bars survive more scales, not more time.
Dark bars exceed the 95th percentile of shuffled maximum lifetimes in that dimension; pale bars do not.
Top six H1 and four H2 bars shown. Core: strong torus features coexist; strict: no extra above-null bars. Neither proves an attractor.